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CO_2 - Assessment Tool- 3 [CSA4305]

Question 1

→ Responsive web Design (RWD)

Responsive web Design (RWD) is a web design approach that makes a website automatically adjust its layout and content according to the screen size of the device. Whether the user opens the website on a mobile phone, tablet, laptop, or desktop, the website should look attractive and easy to use.

For example, an online shopping website should display product in a single column on a mobile app, two columns on a tablet or multiple columns on a desktop.

→ importance of Responsive web Design.

* Provides a better user experience on all devices.
* Eliminates the need for separate mobile and desktop websites.
* Helps improve SEO (Search Engine optimization).
* Reduces maintances cost because only one website is managed.

→ CSS Techniques used for Responsiveness.

1. Flexible layouts

Insted of fixed widths (pixels), use percentages so that elements automatically adjust according to screen size.

Example:

```
.container{
    width:100%;
}
```

2. **Flexible images:**

Images should resize based on the screen size.

Example:

```
img {
    max-width : 100%.
    height : auto ;
}
```

3. **media Queries:**

media queries allow different css styles for different screen size.

Example:

```
@media (max-width : 768PX) {
        body {
            font-size : 16PX;
        }
}
```

→ **using CSS flex box for adaptive Layout:**

flex box helps arrange webpages elements in rows or columns and automatically adjust them based on the available screen space.

Example code:

```
<div class = " container" >
    <div class = "box" > home </div>
    <div class = "box"> About </div>
    <div class = "box" > contact</div>
</div>
```

→ **Desktop:**

Header		
menu	content	slide bar
footer.		

→ Mobile:

Header
menu
content
sidebar
footer.

→ Improvement for Better user Experience.

* Provides a better user experience
* Makes website accessible on all devices.
* improves SEO & increases user engagement & satifaction.
* Reduces maintances by using a single website for all devices.
* use responsive navigation menu for mobile users.
* Test the website on different devices & browser image for faster loading.

Question 2:

→ Client - side validation.

The process of checking user input in the browser before the form is submitted to the server. It helps ensure that users enter valid and complete information.

→ Purpose of client - slide validation:

* Prevents invalid or empty data form being submitted.
* Provides instant feedback to users
* Reduces unnecessary server requests.
* improves user experience by displaying errors immediately.

→ Java script event handling in validation.

Java script uses events such as on submit, oninput, onchange and onclick to validate user input. when the user submits the form, Java script checks whether the entered data is valid. If the data is incorrect, an error message is displayed and the form submission is stopped.

→ Role of Regular expression.

* Email:: checks whether the email follows the correct format (eg; abc@gmail.com).

* Phone number: checks whether the phone number contain exactly 10 digits.

Example:

```
function validate form() {
    let email= document. get element by Id ("email").value
    let email pattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]
              +\.[A-Za-z]{2,}$/;

    if (!emailPattern.test email)) {
            altert ("please enter a valid email address.");
        return false;
    }
    return true;
}
```

→ Improvements for better usability and security.

* Display clear error messages near the input field.

* highlight invalid fields. HTML5 validation along with Javascript.

* use HTTPs to product user data during transmission.

→ Question 3

→ CSS Box Model & Responsive Layout.

When a webpage layout breaks after resizing the browser window, it is usually due to fixed sizes, incorrect positioning or missing responsive design techniques.

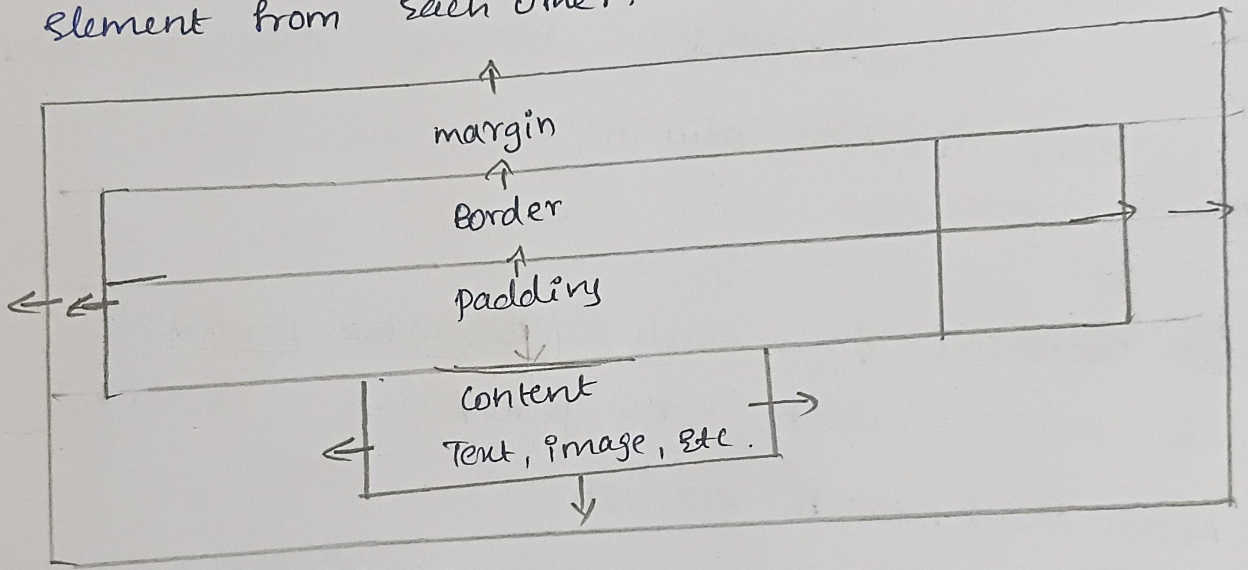
→ CSS Box Model.

1. content - The actual text or image inside the element.

2. Padding - The space between the content and the border.

3. Border - The line that surrounds the padding and content.

4. Margin - The space outside the border that separates element from each other.



→ CSS positioning Techniques.

Static:. Default position

Relative:. Move an element relative to its normal position.

Absolute:. Positions an element relative to its nearest position parent

Fixed:. Keep the element fixed on the screen while scrolling.

Sticky: Acts like relative until scrolling reaches a certain Points.

→ Causes of Layout Breaking.

* fixed width elements (using pixels instead of Percentages)

* Large images that do not resize

* incorrect use of CSS positioning

* Missing media queries

* overflowing content or Long text.

→ using media Queries.

Example:.

```
@media (max-width: 768px) {
        .container {
    flex-direction: column;
      }
}
```

→ Best practerices for a stable & Responsive Layout.

* use flexible Layouts with percentages or flexbox/grid.

* Avoid fixed - width Elements

* Make images responsive using max-width: 100%

* use media queries for different screensize.

* Test the website on mobile, tablet, and desktop devices

* maintain proper spacing, Margins and padding.

Question - 4

→ Document Objects model (DOM)

The Document Object model (DOM) is a programming interface that Represents an HTML webpages as a tree of objects. It allows Java script to access, modify add or remove HTML Element without reloading the Entire webpage.

→ Role of DOM in web Development.

* Represents HTML elements as objects.
* Allows Java script to access and modify webpages content.
* Enables dynamic upadates without refreshing the page
* improve user interaction and responsiveness.
* supports creating interactive web application.

→ Javascript DOM manipulation.

→ getElement By Id ()
→ query selector()
→ inner HTML
→ text content.

Example:

HTML:

<P id = "demo" > welcome! </P>

<button on click = " Chang Text () " > click

Javascript:

function change Text() {
document . get Element By Id("demo").
}

→ Output:

When the user click the click me button the text change from "welcome" to "welcome to JavaScript Dom!" without reloading the webpage.

→ Improvements for Better performance and user Experience.

* update only the required elements instead of reloading the Entire Page.

* Minimize unnecessary Dom manipulations.

* use Event handling Efficiently

* Display Loading indicators for long operations.

* optimize Java script code for faster Performance.

* Test the webpage on different browser and devices.